

RaySection: Executable Section Compression of Calibrated Ray Evidence into Editable CAD

DA3-CAD project

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Abstract

Converting measured geometry into an editable CAD program requires both an approximation model and executable solid topology. We study a training-free compiler from calibrated target masks and first-hit depth to a sequence of planar sketch extrusions. RaySection aggregates silhouette and free-space contradictions on a GPU voxel grid, estimates candidate frames from measured depth, and optimizes contiguous section partitions with an explicit complexity cost. It enumerates the best partitions at successive section budgets and admits only programs that compile to one valid solid with a watertight oriented tessellation. This separates discrete approximation quality from CAD-kernel executability. On 30 rotated, dimension-perturbed procedural instances from ten families, mean shared-frame volume IoU is 0.837, compared with 0.776 for one section and 0.692 for uniform sections. The proposed programs use 1.50 extrusions on average, compared with 8.63 for uniform partitioning. These measurements use simulated noisy depth, not learned depth from photographs. A separate five-case real-RGB audit tests integration with existing COLMAP/PatchMatch outputs and retains source-view abstentions. The release includes a command-line tool, a small redistributable example, fixed-protocol ablations and per-instance evidence. The method is a compact candidate generator, not a recovery of hidden topology, manufacturing tolerances or original design history.

1 Introduction

A surface reconstruction and a CAD reconstruction answer different questions. A surface records observed shape, whereas an editable solid commits to a set of constructive operations, closed topology and geometry in unobserved regions. Point aggregation loses the direction from which each measurement was observed. This matters for cavities: a point inside a coarse solid can represent a visible inner wall, an outlier, or a different surface projected into the same region. Keeping first-hit ray evidence provides an explicit distinction between known free space and space that is merely hidden behind an observation.

We investigate a restricted and reproducible setting: one stationary object, calibrated pinhole cameras, target masks and camera-z depth. Depth can originate from multi-view stereo, but the compiler itself does not recover camera poses, segment the target, or predict depth from RGB. Its output is an editable list of polygonal sketches, planes and extrusion heights, accompanied by a STEP solid. The central engineering issue is that a low-error section approximation can still produce disconnected pieces or a CAD-kernel failure after vectorization.

Our contribution is an executable section-search formulation and an open implementation. A discrete optimizer produces a small pool of budgeted section programs; a bounded CAD compiler filters this pool. We measure the separate effects of depth carving, adaptive partitioning and budget enumeration, and report both successes and failures. We do not claim to introduce space carving, sketch extrusion, dynamic programming, or executed CAD feedback. Nor do the experiments establish superiority over learned CAD systems.

2 Related work

Space carving studies reconstruction through agreement with calibrated photographs [4]. Our occupancy completion uses target silhouettes and observed depth contradictions; it does not implement the full photo-hull theory or inherit its guarantees. The convention that space behind a first hit is unknown is essential to interpreting the resulting solid.

Editable primitive reconstruction is also established. CSG-Stump provides a regular learned constructive representation [7]; CAPRI-Net learns adaptive assemblies of quadric primitives [9]. Lambourne et al. [5]

reconstruct prismatic CAD through profile images and envelope functions, while SECAD-Net learns sketch-extrude assemblies [6]. The recent 3D2EP representation further supports expressive extruded profiles along curved paths [8]. These studies rule out a broad novelty claim for decomposing volumetric geometry into profiles.

MV2Cyl reconstructs extrusion cylinders from multi-view image evidence using learned surface and base-curve information [2]. BrepGaussian learns geometry and patch features from images before parametric fitting [10]. CADENA predicts CAD operations step by step and compares executed intermediate geometry [3]. In contrast, the present compiler uses no learned checkpoint and confines its search to a finite set of parallel section partitions. The quantitative comparisons below are controlled internal ablations, not reruns of these external methods.

3 Method

3.1 Inputs and coordinate contract

Let K_v and $[R_v \mid t_v]$ be the intrinsic and world-to-camera matrices of view v , M_v its binary mask and D_v its first-hit camera-z depth. Zero or nonfinite depth denotes missing evidence. Valid depth pixels are backprojected by

$$p_{vu} = R_v^\top (D_v(u) K_v^{-1} \tilde{u} - t_v). \quad (1)$$

Only fitting-view depths estimate bounds and orientation. We consider the world frame, a PCA frame, and a frame derived from two approximately perpendicular dominant planes. Plane hypotheses use 192 seeded point triples, a distance tolerance of 0.008 times the sampled maximum extent and an inlier-covariance normal refinement. This finite frame set is a practical proposal mechanism, not an exhaustive search over rotations.

For each frame, the 0.2th and 99.8th coordinate percentiles define a robust extent. A padded cubic grid contains N^3 voxel centers, with $N = 72$ in the study. Coordinates remain in the supplied world frame throughout inference and export; no reference mesh determines normalization, pose or scale.

3.2 Ray-consistent completion

Each voxel center is projected into every fitting view. Silhouette sampling uses nearest pixels and a one-pixel dilated mask. A voxel needs visibility in at least half the fitting views, with a minimum of three. It is rejected if more than 5% of its visible projections fall outside the tolerant masks. Valid target depth votes identify free-space contradictions:

$$f_v(p) = \mathbf{1}[z_v(p) < D_v(\pi_v(p)) - 1.5h], \quad (2)$$

where h is the voxel width. We carve a voxel when at least two depth votes contradict it and these exceed 15% of its valid depth votes. The tolerance and vote thresholds are heuristic and fixed across the compared variants. The largest connected voxel component becomes the completion volume V ; discarded components are counted in the report.

A surviving voxel is *not* certified material. In particular, the region behind the first hit remains a completion hypothesis. Silhouette constraints and other camera rays may subsequently reject it, but an unobserved sealed cavity cannot be inferred by this procedure. GPU tensor operations accelerate projection, vote aggregation and the interval statistics below.

3.3 Optimal discrete interval profiles

For a candidate extrusion axis, reorder V into binary slices $V_1, \dots, V_n$. Trim leading and trailing empty slices. An interval $[i, j)$ uses one profile P_{ij} repeated over its length $\ell = j - i$. At profile pixel u , let $c_{ij}(u) = \sum_{s=i}^{j-1} V_s(u)$. The profile minimizing voxel XOR error is

$$P_{ij}(u) = \mathbf{1}[c_{ij}(u) \geq \ell/2], \quad E(i, j) = \frac{\sum_u \min(c_{ij}(u), \ell - c_{ij}(u))}{\max(1, |V|)}. \quad (3)$$

This follows because the two choices, empty and occupied, incur $c_{ij}(u)$ and $\ell - c_{ij}(u)$ disagreements independently at each pixel. Axial prefix sums provide all interval counts without repeatedly summing slices.

For a partition $\mathcal{P}$, minimize

$$J(\mathcal{P}) = \sum_{[i,j) \in \mathcal{P}} E(i, j) + \lambda |\mathcal{P}|, \quad |\mathcal{P}| \leq B, \quad (4)$$

with $\lambda = 0.03$ and $B = 8$. The recurrence

$$F(k, j) = \min_{i < j} \{F(k-1, i) + E(i, j) + \lambda\}, \quad F(0, 0) = 0 \quad (5)$$

gives the exact optimum for the fixed discrete volume, frame, axis and section budget. Interval construction is $O(N^4)$ for a cubic grid and the dynamic program is $O(BN^2)$ after costs are available. Implementation constants and CAD compilation dominate the small-grid experiments.

3.4 Executable budget search

The discrete optimum need not be an executable one-solid CAD program. Contour simplification can remove a narrow connection; separated profile components can remain disconnected after extrusion. We therefore retain the distinct best partitions obtained with budgets $1, \dots, B$ across three axes and the available frames, sort them by J , and attempt compilation in order. This pool contains at most $9B$ candidates. It is a bounded heuristic for the kernel-constrained problem: a second-best partition at a fixed budget is not necessarily in the pool, so global optimality over executable programs is not claimed.

Contours are simplified with a 0.65-pixel tolerance, retaining exterior loops and holes. Each connected profile becomes a planar sketch extrusion. Adjacent sections overlap by $0.001h$ to reduce exact-contact numerical failures. OpenCascade unions the pieces; admission requires a valid positive-volume single solid and a watertight, consistently oriented tessellation. Each attempt runs in an isolated process with a 20-second timeout. Failed attempts and the selected objective are retained. Runtime-dependent timeout outcomes are not a bitwise determinism guarantee. The section cap differs from the emitted extrusion count when a profile contains several components.

4 Experimental protocol

4.1 Controlled geometry study

Three seed-42 instances (block, hexagonal prism and L-bracket) supported initial development. We then generated 30 instances with seed 20260910, three from each of ten existing procedural families. Each scalar dimension is independently multiplied by a uniform factor in $[0.85, 1.15]$. A random axis and an angle in $[15^\circ, 165^\circ]$ rotate each shape. Families include convex prisms, L/T/U profiles, single- and multiple-hole plates, a flange, a stepped shaft and a bottle-like stepped body. These are related procedural families, not a representative sample of manufactured objects or unseen CAD categories.

Sixteen calibrated 128×128 views use an azimuth increment of 137.507764° and four repeated elevations from -55° to 55° . Ray casting generates masks and first-hit z depth. Gaussian depth noise has standard deviation 0.001 times object extent, and 10% of depth pixels are randomly missing. Every fourth view is excluded from fitting, leaving twelve fitting and four evaluation views. This deterministic split reserves the highest elevation ring; it is not random pixel holdout. Masks and cameras are exact simulation outputs. No RGB-to-depth network is evaluated in this experiment.

All fitting routines receive only ray bundles. Reference meshes are read after STEP generation. Volume IoU is measured in the shared observation frame. For squared symmetric Chamfer, both meshes receive the *same* reference-defined translation and isotropic scale after reconstruction; 8192 area-weighted points are sampled per mesh. No prediction-specific centering, scaling or ICP is used. Invalid outputs contribute zero IoU. Other metrics are conditional on valid outputs, with the validity denominator displayed.

We compare a single section, eight uniform axial intervals, the adaptive compiler without depth carving, adaptive partitioning without budget enumeration, and full RaySection. Even the no-depth-carving ablation shares measured depth-derived bounds and frames. All variants share the same mask tolerances, frame proposals, contour simplification and kernel admission checks. Budget enumeration was added after exploratory real-case compilation failures; the grid, carving rules and penalty were unchanged. Both versions and failures are retained. This is a development-informed controlled study, not a blinded confirmatory trial.

4.2 Exploratory real-RGB audit

We reuse five previously inspected T-LESS instances [1], each with 32 RGB views. Existing RGB-only COLMAP poses and masked PatchMatch depths provide inputs; BOP instance masks and fixed instance selection are oracle inputs. No sensor depth or reference CAD enters fitting. The new compiler uses 24 views. Since upstream SfM/MVS used the full pool, the remaining eight views are held out only from the CAD fitter, not from the complete image-processing pipeline. The historical v9 system fitted all 32 views and is a diagnostic comparison, not an equal-input baseline.

Method	Valid	IoU $\uparrow$	CD ² $\downarrow$	View IoU	Extr.	Time (s)
Single section	30/30	0.776	2.64	0.864	1.00	1.85
Uniform sections	27/30	0.692	8.22	0.854	8.63	8.83
No depth carving	30/30	0.647	3.33	0.842	2.27	2.13
No budget search	29/30	0.818	0.65	0.913	2.17	1.87
RaySection	30/30	0.837	0.71	0.917	1.50	2.10

Table 1: Controlled simulated-depth study, 30 instances. CD² is symmetric squared Chamfer times 1000; view IoU uses fitting-excluded masks. IoU includes invalid outputs as zero. Remaining means and median fit time condition on valid outputs. Extr. counts emitted extrusion solids, not section intervals.

Source verification uses the original RGB/mask/depth evidence at maximum image dimension 128. We retain the existing thresholds: silhouette IoU 0.87, depth inlier fraction 0.90, appearance-edge precision 0.45 and recall 0.12, with the edge gate active when at least 128 observed edge pixels are available. These are provisional engineering gates, not calibrated probabilities of correct reconstruction. We do not use the earlier independently normalized reference IoU: an audit found unresolved rotation between reconstruction and reference frames, invalidating its interpretation as aligned shape accuracy.

5 Results

Table 1 separates reconstruction quality from program size. RaySection achieves mean IoU 0.837 with 1.50 extrusions, while uniform partitioning achieves 0.692 with 8.63 extrusions. The one-section baseline reaches 0.776; removing depth carving gives 0.647. The paired effect estimates and both instance- and family-level bootstrap intervals are supplied with the result ledger. Because three variants come from each family, the family-resampled interval is the more conservative check on dependence. We report descriptive intervals rather than treating the thousands of pixels or sampled surface points as independent experimental units.

Comparison	Mean IoU difference	95% family bootstrap interval
RaySection minus Single section	+0.061	[+0.011, +0.127]
RaySection minus Uniform sections	+0.145	[+0.008, +0.300]
RaySection minus No depth carving	+0.189	[+0.137, +0.247]
RaySection minus No budget search	+0.019	[-0.015, +0.075]

Table 2: Paired differences. Percentile intervals resample the ten family means 10,000 times. They are descriptive and not adjusted for multiple comparisons.

The family-resampled interval for the gain over one section is positive on this procedural study, but the interval for budget search versus the best-partition-only variant crosses zero. Budget search raises validity from 29/30 to 30/30 and reduces mean program size; its mean conditional Chamfer is slightly worse (0.71 versus 0.65). Thus its clearest measured benefit is executability and compactness, not a uniform geometric improvement. Two instances lose IoU relative to the single-section baseline, and eleven lose IoU relative to uniform partitioning. These regressions are retained in the ledger.

Executable budget search produced 5/5 kernel-valid candidates, compared with 2/5 for the best-partition-only variant. Only 0/5 candidates passed every existing source-view gate. Table 3 therefore supports an integration and executability result, not reliable real-photo CAD recovery. The historical v9 comparison also remains an exploratory diagnostic. A lower-complexity fallback can regain a valid solid while losing geometric detail; validity is not a monotonic accuracy guarantee.

6 Limitations and reproducibility

The representation is a stack of parallel prismatic sections in a selected frame. Smooth curvature becomes polygonal or stepped; oblique interacting features can require many sections. The procedure produces editable coordinates and heights, not semantic constraints, circles, fillets, tolerances or a recovered designer feature tree. Its finite frame proposals and largest-component filter can discard real structure. Ray thresholds assume reasonably calibrated masks and depth; transparency, specularity, motion and inconsistent MVS can invalidate those assumptions. Hidden cavities remain unidentifiable without observations.

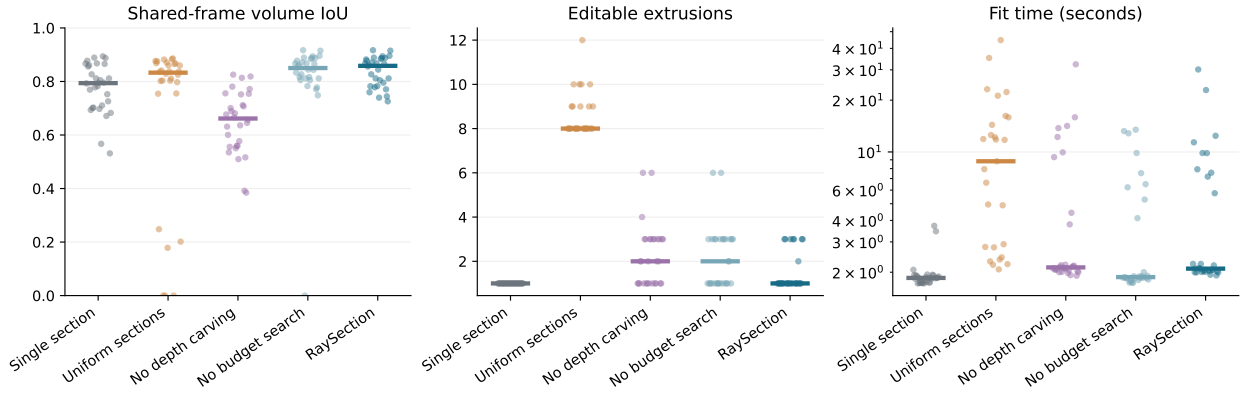


Figure 1: Per-instance controlled outcomes. Points are individual procedural instances; horizontal marks show medians. Time includes isolated CAD compilation and failed candidate attempts, not data generation. Log time scale.

Case	v9 view IoU	Ours view IoU	Ours depth	Ours edge P	Decision
o02-fixed	0.891	0.875	0.971	0.284	ABSTAIN
o04-fixed	0.862	0.896	0.987	0.291	ABSTAIN
o10	0.680	0.334	0.370	0.447	ABSTAIN
o20-fixed	0.755	0.413	0.472	0.416	ABSTAIN
o25	0.880	0.839	0.904	0.354	ABSTAIN

Table 3: Exploratory real RGB. View IoU and depth use the eight views excluded from the new fitter; edge precision and final decisions use the full verifier. The historical system used all views during fitting. These values measure source agreement, not reference-CAD accuracy or hidden-topology correctness.

The controlled data favor prismatic and stepped families. Exact synthetic masks and calibration simplify the task. Thirty instances from ten generators are too few to establish broad industrial generalization. The real cases are exploratory and use oracle target selection. Learned competitors were not rerun. Novelty claims are limited to the specified implementation and formulation; a focused literature search cannot establish priority for every component combination.

Code, scripts, source hashes, per-instance ledgers and a generated ray bundle are released with the DA3-CAD repository. The CLI supports CPU and CUDA and refuses to overwrite an existing result directory. The measured environment is Python 3.12, PyTorch 2.13 with CUDA 13 and an RTX 5080 with 16 GB memory. Some study jobs shared the workstation, so the recorded times are descriptive end-to-end measurements rather than isolated hardware throughput benchmarks. The release separates candidate generation from downstream acceptance. No third-party model weights or T-LESS imagery are redistributed.

AI assistance and review. An OpenAI coding assistant contributed algorithm exploration, implementation, experiment execution, literature lookup, drafting and internal consistency review under the project maintainer’s instructions. Measurements were computed by the released scripts. Internal critique was same-model self-review, not independent peer review. The maintainer must confirm authorship and responsibility before external submission. This report is not a claim of publication acceptance.

7 Conclusion

RaySection provides a training-free path from calibrated ray evidence to compact, executable sketch-extrusion candidates. Exact interval optimization addresses a restricted approximation problem; budget enumeration and CAD admission address the separate problem of executable topology. The controlled study quantifies the resulting accuracy–complexity trade-off, while the real-RGB audit preserves failures and abstentions. This separation supports reproducible method development without equating a valid STEP file with a correct reconstruction.

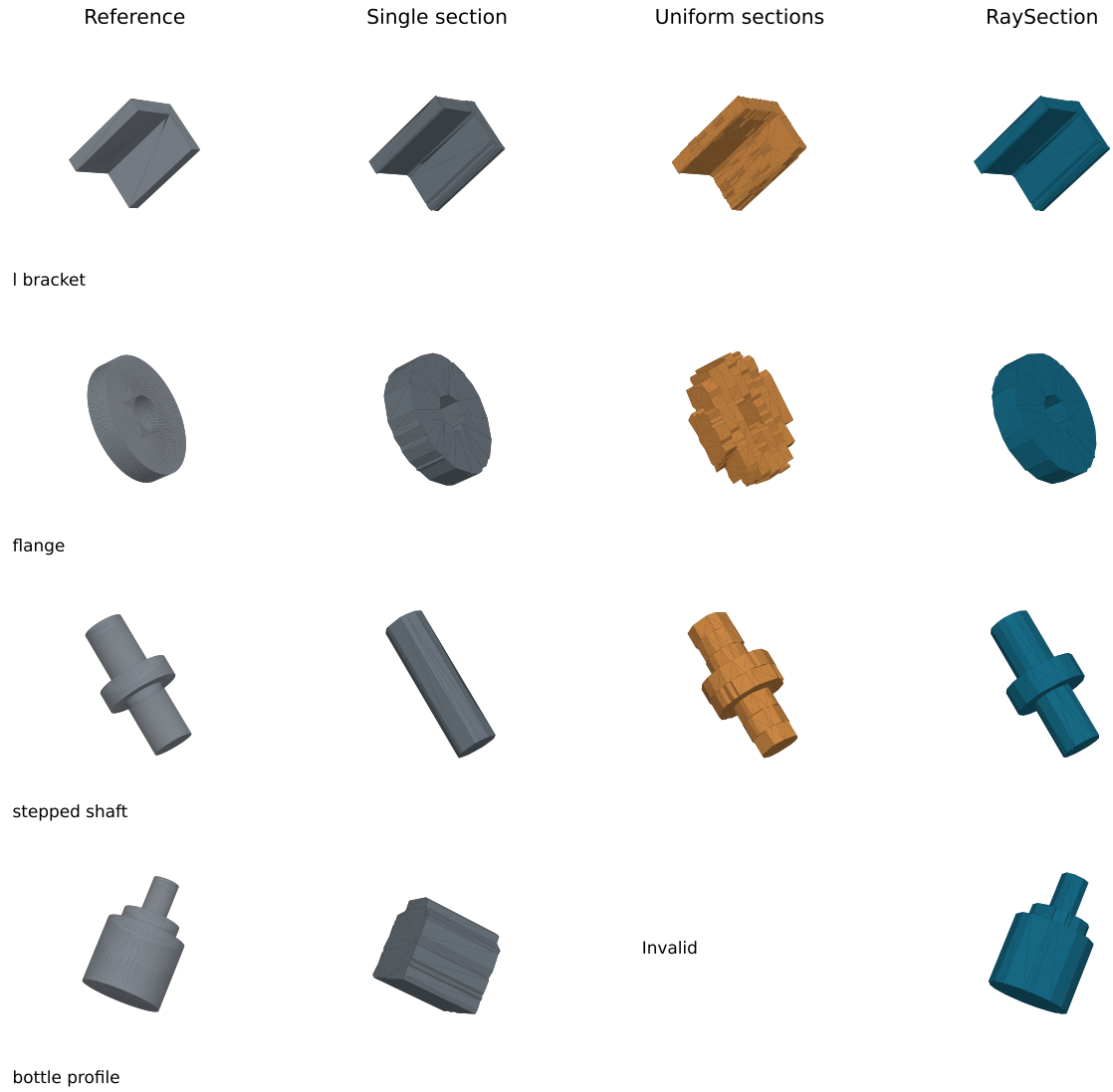


Figure 2: Shared-frame qualitative comparison on the first generated L-bracket, flange, stepped-shaft and bottle-profile instances. Rows were selected by family and generation index, not by measured improvement. Polygonal profiles and stair-step approximations remain visible limitations.

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